
You Only Look at One Sequence: Rethinking Transformer in Vision through Object Detection

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Abstract

Can Transformer perform 2D object- and region-level recognition from a pure sequence-to-sequence perspective with minimal knowledge about the 2D spatial structure? To answer this question, we present You Only Look at One Sequence (YOLOS), a series of object detection models based on the vanilla Vision Transformer with the fewest possible modifications, region priors, as well as inductive biases of the target task. We find that YOLOS pre-trained on the mid-sized ImageNet-1k dataset *only* can already achieve quite competitive performance on the challenging COCO object detection benchmark, *e.g.*, YOLOS-Base directly adopted from BERT-Base architecture can obtain 42.0 box AP on COCO val. We also discuss the impacts as well as limitations of current pre-train schemes and model scaling strategies for Transformer in vision through YOLOS. Code and pre-trained models are available at <https://github.com/hustvl/YOLOS>.

1 Introduction

Transformer [58] is born to transfer. In natural language processing (NLP), the dominant approach is to first pre-train Transformer on large, generic corpora for general language representation learning, and then fine-tune or adapt the model on specific target tasks [18]. Recently, Vision Transformer (ViT) ¹ [21] demonstrates that canonical Transformer encoder architecture directly inherited from NLP can perform surprisingly well on image recognition at scale using modern vision transfer learning recipe [33]. Taking sequences of image patch embeddings as inputs, ViT can successfully transfer pre-trained general visual representations from sufficient scale to more specific image classification tasks with fewer data points from a pure sequence-to-sequence perspective.

Since a pre-trained Transformer can be successfully fine-tuned on sentence-level tasks [7, 19] in NLP, as well as *token-level* tasks [48, 52], where models are required to produce fine-grained output at the token-level [18]. A natural question is: Can ViT transfer to more challenging *object- and region-level* target tasks in computer vision such as object detection other than image-level recognition?

ViT-FRCNN [6] is the first to use a pre-trained ViT as the backbone for a Faster R-CNN [50] object detector. However, this design cannot get rid of the reliance on convolutional neural networks (CNNs)

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¹There are various sophisticated or hybrid architectures termed as “Vision Transformer”. For disambiguation, in this paper, “Vision Transformer” and “ViT” refer to the canonical or vanilla Vision Transformer architecture proposed by Dosovitskiy et al. [21] unless specified.

你只看一个序列：通过目标检测重新思考视觉中的Transformer

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摘要

Transformer能否以纯序列到序列的视角，在仅具备最少二维空间结构知识的情况下完成二维物体及区域级识别？为回答这一问题，我们提出了“仅需观察一个序列”（YOLOS）——这一系列基于原始视觉Transformer的目标检测模型仅需极少的修改、区域先验知识以及目标任务的归纳偏置。我们发现，仅通过中等规模的ImageNet-1k数据集*only*进行预训练的YOLOS，已在极具挑战性的COCO目标检测基准上取得相当竞争力的性能：*e.g.* 直接沿用BERT-Base架构的YOLOS-Base可在COCO验证集上获得42.0的边界框平均精度。我们还通过YOLOS探讨了当前预训练方案及Transformer视觉模型缩放策略的影响与局限性。代码与预训练模型已发布于<https://github.com/hustv1/YOLOS>。

1 引言

Transformer [58] 生来就是为了迁移。在自然语言处理（NLP）领域，主流方法是先在大型通用语料库上对Transformer进行预训练，以学习通用的语言表示，然后在特定的目标任务上对模型进行微调或适配[18]。最近，Vision Transformer (ViT) ¹ [21] 证明，直接继承自NLP的经典Transformer编码器架构，采用现代视觉迁移学习方案[33]，在大规模图像识别任务上能表现出惊人的性能。ViT以图像块嵌入序列作为输入，从纯粹的序列到序列视角出发，能够成功地将预训练的通用视觉表示从足够大的规模迁移到数据点较少的、更具体的图像分类任务中。

由于预训练的Transformer可以在NLP中的句子级任务[7, 19]以及*token-level*任务[48, 52]上成功进行微调，这些任务要求模型在标记级别生成细粒度输出[18]。一个自然的问题是：ViT能否迁移到计算机视觉中更具挑战性的*object- and region-level*目标任务，例如图像级识别之外的目标检测？

ViT-FRCNN [6] 首次采用预训练的ViT作为Faster R-CNN [50]目标检测器的主干网络。然而，该设计仍无法摆脱对卷积神经网络（CNNs）的依赖。

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and strong 2D inductive biases, as ViT-FRCNN re-interprets the output sequences of ViT to 2D spatial feature maps and depends on region-wise pooling operations (*i.e.*, RoIPool [23, 25] or RoIAlign [27]) as well as region-based CNN architectures [50] to decode ViT features for object- and region-level perception. Inspired by modern CNN design, some recent works [39, 59, 62, 65] introduce the pyramidal feature hierarchy, spatial locality, equivariant as well as invariant representations [24] to canonical Vision Transformer design, which largely boost the performance in dense prediction tasks including object detection. However, these architectures are performance-oriented and cannot reflect the properties of the canonical or vanilla Vision Transformer [21] directly inherited from Vaswani et al. [58]. Another series of work, the DETection TRansformer (DETR) families [10, 72], use a random initialized Transformer to encode & decode CNN features for object detection, which does not reveal the transferability of a pre-trained Transformer.

Intuitively, ViT is designed to model long-range dependencies and global contextual information instead of local and region-level relations. Moreover, ViT lacks hierarchical architecture as modern CNNs [26, 35, 53] to handle the large variations in the scale of visual entities [1, 37]. Based on the available evidence, it is still unclear whether a pure ViT can transfer pre-trained general visual representations from image-level recognition to the much more complicated 2D object detection task.

To answer this question, we present You Only Look at One Sequence (YOLOS), a series of object detection models based on the canonical ViT architecture with the fewest possible modifications, region priors, as well as inductive biases of the target task injected. Essentially, the change from a pre-trained ViT to a YOLOS detector is embarrassingly simple: (1) YOLOS replaces one [CLS] token for image classification in ViT with one hundred [DET] tokens for object detection. (2) YOLOS replaces the image classification loss in ViT with the bipartite matching loss to perform object detection in a set prediction manner following Carion et al. [10], which can avoid re-interpreting the output sequences of ViT to 2D feature maps as well as prevent manually injecting heuristics and prior knowledge of object 2D spatial structure during label assignment [71]. Moreover, the prediction head of YOLOS can get rid of complex and diverse designs, which is as compact as a classification layer.

Directly inherited from ViT [21], YOLOS is not designed to be yet another high-performance object detector, but to unveil the versatility and transferability of pre-trained canonical Transformer from image recognition to the more challenging object detection task. Concretely, our main contributions are summarized as follows:

- We use the mid-sized ImageNet-1k [51] as the *sole* pre-training dataset, and show that a vanilla ViT [21] can be successfully transferred to perform the complex object detection task and produce competitive results on COCO [36] benchmark with the fewest possible modifications, *i.e.*, by only looking at one sequence (YOLOS).
- For the first time, we demonstrate that 2D object detection can be accomplished in a pure sequence-to-sequence manner by taking a sequence of fixed-sized non-overlapping image patches as input. Among existing object detectors, YOLOS utilizes the minimal 2D inductive biases.
- For the vanilla ViT, we find the object detection results are quite sensitive to the pre-train scheme and the detection performance is far from saturating. Therefore the proposed YOLOS can be also used as a challenging benchmark task to evaluate different (label-supervised and self-supervised) pre-training strategies for ViT.

2 You Only Look at One Sequence

As for the model design, YOLOS closely follows the original ViT architecture [21], and is optimized for object detection in the same vein as Carion et al. [10]. YOLOS can be easily adapted to various canonical Transformer architectures available in NLP as well as in computer vision. This intentionally simple setup is not designed for better detection performance, but to exactly reveal characteristics of the Transformer family in object detection as unbiased as possible.

2.1 Architecture

An overview of the model is depicted in Fig. 1. Essentially, the change from a ViT to a YOLOS detector is simple: (1) YOLOS drops the [CLS] token for image classification and appends one

以及强大的二维归纳偏置，因为ViT-FRCNN将ViT的输出序列重新解释为二维空间特征图，并依赖于区域池化操作（*i.e.*, RoIPool [23, 25] 或 RoIAlign [27]）以及基于区域的CNN架构[50]来解码ViT特征，以实现物体和区域级别的感知。受现代CNN设计的启发，近期一些研究[39, 59, 62, 65]将金字塔特征层次、空间局部性、等变及不变表示[24]引入到经典的Vision Transformer设计中，这极大地提升了包括物体检测在内的密集预测任务的性能。然而，这些架构以性能为导向，无法直接反映经典或原始Vision Transformer[21]从Vaswani等人[58]继承而来的特性。另一系列工作，即DEtection TRansformer（DETR）家族[10, 72]，使用随机初始化的Transformer对CNN特征进行编码和解码以进行物体检测，这并未揭示预训练Transformer的可迁移性。

直观上，ViT旨在建模长距离依赖和全局上下文信息，而非局部和区域层面的关系。此外，ViT缺乏现代CNN[26, 35, 53]那样的分层架构来处理视觉实体尺度上的巨大差异[1, 37]。基于现有证据，目前尚不清楚纯ViT能否将预训练的通用视觉表示从图像级识别迁移到更为复杂的2D目标检测任务中。

为回答这一问题，我们提出了“只看一个序列”（YOLOS）模型系列。该系列目标检测模型基于标准ViT架构，仅进行最少量修改，不引入区域先验信息，同时避免注入目标任务的归纳偏置。本质上，从预训练ViT到YOLOS检测器的转变异常简单：（1）YOLOS将ViT中用于图像分类的一个[CLS]标记替换为用于目标检测的一百个[DET]标记。（2）YOLOS采用Carion等人[10]的集合预测方式，将ViT的图像分类损失替换为二分图匹配损失进行目标检测。这种方法既无需将ViT输出序列重新解释为二维特征图，也能在标签分配过程中避免手动注入启发式规则和物体二维空间结构的先验知识[71]。此外，YOLOS的预测头无需复杂多样的设计，其紧凑程度可媲美分类层。

直接继承自ViT[21]，YOLOS并非旨在成为另一个高性能物体检测器，而是为了揭示预训练标准Transformer从图像识别到更具挑战性的物体检测任务中的多功能性和可迁移性。具体而言，我们的主要贡献总结如下：

- 我们使用中等规模的ImageNet-1k [51]作为*sole*预训练数据集，并证明一个普通的ViT [21]可以通过最少的修改成功迁移到复杂的物体检测任务中，在COCO [36]基准测试中产生有竞争力的结果，*i.e.*，仅通过观察一个序列（YOLOS）。
- 我们首次证明，通过将固定尺寸的非重叠图像块序列作为输入，二维物体检测可以以纯粹的序列到序列方式完成。在现有物体检测器中，YOLOS利用了最少的二维归纳偏置。
- 对于原始的ViT，我们发现目标检测结果对预训练方案相当敏感，且检测性能远未饱和。因此，提出的YOLOS也可作为一个具有挑战性的基准任务，用于评估ViT的不同（标签监督和自监督）预训练策略。

2 你只看一个序列

在模型设计方面，YOLOS严格遵循原始ViT架构[21]，并采用与Carion等人[10]相同的思路针对目标检测任务进行优化。YOLOS能够轻松适配自然语言处理与计算机视觉领域中各类经典Transformer架构。这种刻意简化的设计并非为了追求更优的检测性能，而是为了尽可能无偏差地揭示Transformer系列模型在目标检测任务中的本质特性。

2.1 架构

模型概览如图1所示。本质上，从ViT到YOLOS检测器的转变非常简单：（1）YOLOS舍弃了用于图像分类的[CLS]标记，并新增一个

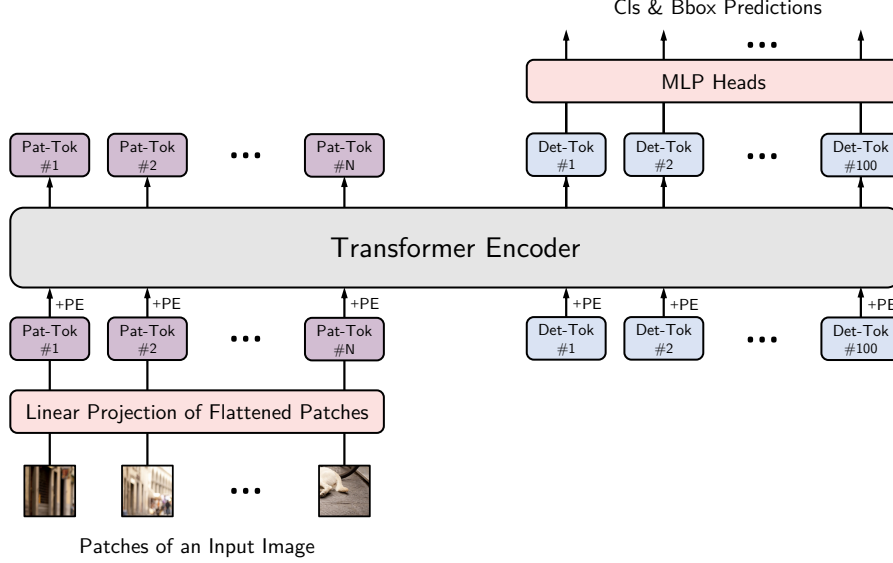


Figure 1: YOLOS architecture overview. “Pat-Tok” refers to [PATCH] token, which is the embedding of a flattened image patch. “Det-Tok” refers to [DET] token, which is a learnable embedding for object binding. “PE” refers to positional embedding. During training, YOLOS produces an optimal bipartite matching between predictions from one hundred [DET] tokens and ground truth objects. During inference, YOLOS directly outputs the final set of predictions in parallel. The figure style is inspired by Dosovitskiy et al. [21].

hundred randomly initialized learnable detection tokens ([DET] tokens) to the input patch embeddings ([PATCH] tokens) for object detection. (2) During training, YOLOS replaces the image classification loss in ViT with the bipartite matching loss to perform object detection in a set prediction manner following Carion et al. [10].

Stem. The canonical ViT [21] receives an 1D sequence of embedded tokens as the input. To handle 2D image inputs, we reshape the image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ into a sequence of flattened 2D image patches $\mathbf{x}_{\text{PATCH}} \in \mathbb{R}^{N \times (P^2 \cdot C)}$. Here, (H, W) is the resolution of the input image, C is the number of input channels, (P, P) is the resolution of each image patch, and $N = \frac{HW}{P^2}$ is the resulting number of patches. Then we map $\mathbf{x}_{\text{PATCH}}$ to D dimensions with a trainable linear projection $\mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}$. We refer to the output of this projection $\mathbf{x}_{\text{PATCH}}\mathbf{E}$ as [PATCH] tokens. Meanwhile, one hundred randomly initialized learnable [DET] tokens $\mathbf{x}_{\text{DET}} \in \mathbb{R}^{100 \times D}$ are appended to the [PATCH] tokens. Position embeddings $\mathbf{P} \in \mathbb{R}^{(N+100) \times D}$ are added to all the input tokens to retain positional information. We use the standard learnable 1D position embeddings following Dosovitskiy et al. [21]. The resulting sequence $\mathbf{z}_0$ serves as the input of YOLOS Transformer encoder. Formally:

$$\mathbf{z}_0 = [\mathbf{x}_{\text{PATCH}}^1\mathbf{E}; \dots; \mathbf{x}_{\text{PATCH}}^N\mathbf{E}; \mathbf{x}_{\text{DET}}^1; \dots; \mathbf{x}_{\text{DET}}^{100}] + \mathbf{P}. \quad (1)$$

Body. The body of YOLOS is basically the same as ViT, which consists of a stack of Transformer encoder layers only [58]. [PATCH] tokens and [DET] tokens are treated equally and they perform global interactions inside Transformer encoder layers.

Each Transformer encoder layer consists of one multi-head self-attention (MSA) block and one MLP block. LayerNorm (LN) [2] is applied before every block, and residual connections [26] are applied after every block [3, 61]. The MLP contains one hidden layer with an intermediate GELU [29] non-linearity activation function. Formally, for the ℓ -th YOLOS Transformer encoder layer:

$$\begin{aligned} \mathbf{z}'_{\ell} &= \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \\ \mathbf{z}_{\ell} &= \text{MLP}(\text{LN}(\mathbf{z}'_{\ell})) + \mathbf{z}'_{\ell}. \end{aligned} \quad (2)$$

Detector Heads. The detector head of YOLOS gets rid of complex and heavy designs, and is as neat as the image classification layer of ViT. Both the classification and the bounding box regression heads are implemented by one MLP with separate parameters containing two hidden layers with intermediate ReLU [41] non-linearity activation functions.

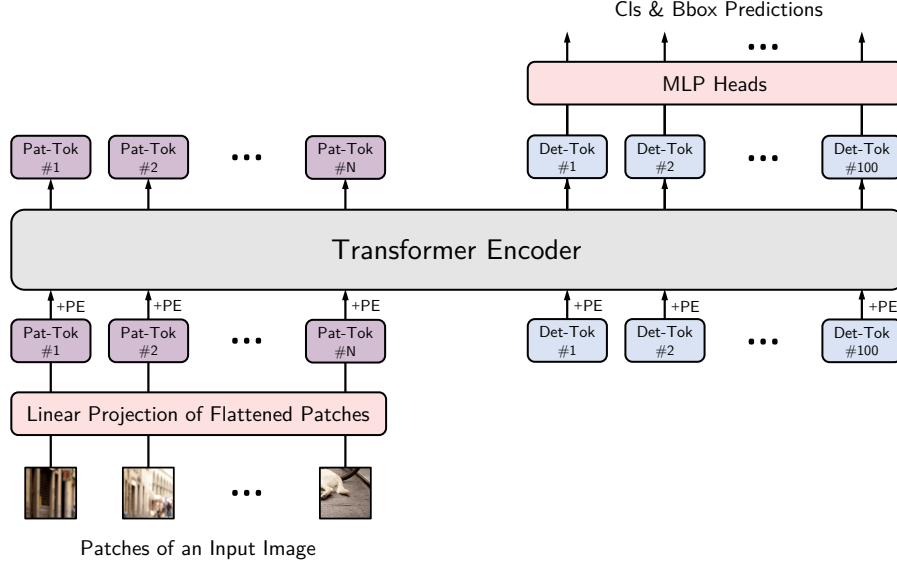


图1: YOLOS架构概览。"Pat-Tok"指[PATCH]标记, 即展平图像块的嵌入表示。"Det-Tok"指[DET]标记, 这是用于目标绑定的可学习嵌入。"PE"指位置嵌入。训练时, YOLOS通过一百个[DET]标记的预测结果与真实目标之间建立最优二分匹配。推理时, YOLOS直接并行输出最终的预测集合。本图风格受Dosovitskiy等人[21]的启发。

在输入补丁嵌入 ([PATCH] 标记) 中添加一百个随机初始化的可学习检测标记 ([DET] 标记) 以进行目标检测。(2) 在训练过程中, YOLOS 遵循 Carion 等人 [10] 的方法, 将 ViT 中的图像分类损失替换为二分图匹配损失, 以集合预测的方式执行目标检测。

主干网络。经典的ViT [21]接收一个一维的嵌入令牌序列作为输入。为处理二维图像输入, 我们将图像 $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ 重塑为一系列展平的二维图像块 $\mathbf{x}_{\text{PATCH}} \in \mathbb{R}^{N \times (P^2 \cdot C)}$ 。其中 (H, W) 是输入图像的分辨率, C 是输入通道数, (P, P) 是每个图像块的分辨率, $N = \frac{HW}{P^2}$ 是最终得到的图像块数量。随后我们通过可训练的线性投影 $\mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}$ 将 $\mathbf{x}_{\text{PATCH}}$ 映射到 D 维。我们将该投影的输出 $\mathbf{x}_{\text{PATCH}}\mathbf{E}$ 称为[PATCH]令牌。同时, 一百个随机初始化的可学习[DET]令牌 $\mathbf{x}_{\text{DET}} \in \mathbb{R}^{100 \times D}$ 被附加到[PATCH]令牌之后。为保留位置信息, 所有输入令牌都会加入位置嵌入 $\mathbf{P} \in \mathbb{R}^{(N+100) \times D}$ 。我们采用Dosovitskiy等人[21]提出的标准可学习一维位置嵌入方法。最终得到的序列 $\mathbf{z}_0$ 将作为YOLOS Transformer编码器的输入。形式化表示为:

$$\mathbf{z}_0 = [\mathbf{x}_{\text{PATCH}}^1 \mathbf{E}; \dots; \mathbf{x}_{\text{PATCH}}^N \mathbf{E}; \mathbf{x}_{\text{DET}}^1; \dots; \mathbf{x}_{\text{DET}}^{100}] + \mathbf{P}. \quad (1)$$

主体。YOLOS的主体与ViT基本相同, 仅由一系列Transformer编码器层构成[58]。PATCH标记和DET标记被同等对待, 它们在Transformer编码器层内部进行全局交互。

每个Transformer编码器层由一个多头自注意力 (MSA) 模块和一个MLP模块组成。在每个模块之前应用LayerNorm (LN) [2], 在每个模块之后应用残差连接[26][3, 61]。MLP包含一个具有中间GELU[29]非线性激活函数的隐藏层。形式上, 对于 ℓ -th YOLOS Transformer编码器层:

$$\begin{aligned} \mathbf{z}'_{\ell} &= \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \\ \mathbf{z}_{\ell} &= \text{MLP}(\text{LN}(\mathbf{z}'_{\ell})) + \mathbf{z}'_{\ell}. \end{aligned} \quad (2)$$

检测头。YOLOS的检测头摒弃了复杂而笨重的设计, 简洁程度堪比ViT的图像分类层。分类和边界框回归头均由一个多层感知机实现, 该感知机拥有独立的参数, 包含两个隐藏层, 中间采用ReLU[41]非线性激活函数。

Detection Token. We purposefully choose randomly initialized [DET] tokens as proxies for object representations to avoid inductive biases of 2D structure and prior knowledge about the task injected during label assignment. When fine-tuning on COCO, for each forward pass, an optimal bipartite matching between predictions generated by [DET] tokens and ground truth objects is established. This procedure plays the same role as label assignment [10, 71], but is unaware of the input 2D structure, *i.e.*, YOLOS does not need to re-interpret the output sequence of ViT to an 2D feature maps for label assignment. Theoretically, it is feasible for YOLOS to perform any dimensional object detection without knowing the exact spatial structure and geometry, as long as the input is always flattened to a sequence in the same way for each pass.

Fine-tuning at Higher Resolution. When fine-tuning on COCO, all the parameters are initialized from ImageNet-1k pre-trained weights except for the MLP heads for classification & bounding box regression as well as one hundred [DET] tokens, which are randomly initialized. During fine-tuning, the image has a much higher resolution than pre-training. We keep the patch size P unchanged, *i.e.*, $P \times P = 16 \times 16$, which results in a larger effective sequence length. While ViT can handle arbitrary input sequence lengths, the positional embeddings need to adapt to the longer input sequences with various lengths. We perform 2D interpolation of the pre-trained position embeddings on the fly².

Inductive Bias. We carefully design the YOLOS architecture for the minimal additional inductive biases injection. The inductive biases inherent from ViT come from the patch extraction at the network stem part as well as the resolution adjustment for position embeddings [21]. Apart from that, YOLOS adds no non-degenerated (*e.g.*, 3×3 or other non 1×1) convolutions upon ViT³. From the representation learning perspective, we choose to use [DET] tokens to bind objects for final predictions to avoid additional 2D inductive biases as well as task-specific heuristics. The performance-oriented design inspired by modern CNN architectures such as pyramidal feature hierarchy, 2D local spatial attention as well as the region-wise pooling operation is not applied. All these efforts are meant to exactly unveil the versatility and transferability of pre-trained Transformers from image recognition to object detection in a pure sequence-to-sequence manner, with minimal knowledge about the input spatial structure and geometry.

Comparisons with DETR. The design of YOLOS is deeply inspired by DETR [10]: YOLOS uses [DET] tokens following DETR as proxies for object representations to avoid inductive biases about 2D structures and prior knowledge about the task injected during label assignment, and YOLOS is optimized similarly as DETR.

Meanwhile, there are some key differences between the two models: (1) DETR adopts a Transformer encoder-decoder architecture, while YOLOS chooses an encoder-only Transformer architecture. (2) DETR only employs pre-training on its CNN backbone but leaves the Transformer encoder & decoder being trained from random initialization, while YOLOS naturally inherits representations from any pre-trained canonical ViT. (3) DETR applies cross-attention between encoded image features and object queries with auxiliary decoding losses deeply supervised at each decoder layer, while YOLOS always looks at only one sequence for each encoder layer, without distinguishing [PATCH] tokens and [DET] tokens in terms of operations. Quantitative comparisons between the two are in Sec. 3.4.

3 Experiments

3.1 Setup

Pre-training. We pre-train all YOLOS / ViT models on ImageNet-1k [51] dataset using the data-efficient training strategy suggested by Touvron et al. [57]. The parameters are initialized with a truncated normal distribution and optimized using AdamW [40]. The learning rate and batch size are 1×10^{-3} and 1024, respectively. The learning rate decay is cosine and the weight decay is 0.05. Rand-Augment [14] and random erasing [69] implemented by `timm` library [64] are used for data augmentation. Stochastic depth [32], Mixup [68] and Cutmix [66] are used for regularization.

²The configurations of position embeddings are detailed in the Appendix.

³We argue that it is imprecise to say Transformer do not have convolutions. All linear projection layers in Transformer are equivalent to point-wise or 1×1 convolutions with sparse connectivity, parameter sharing, and equivalent representations properties, which can largely improve the computational efficiency compared with the “all-to-all” interactions in fully-connected design that has even weaker inductive biases [5, 24].

检测令牌。我们特意选择随机初始化的[DET]令牌作为物体表征的代理，以避免标签分配过程中引入的二维结构归纳偏倚和任务先验知识。在COCO数据集上进行微调时，每次前向传播都会在[DET]令牌生成的预测结果与真实物体之间建立最优二分匹配。这一过程与标签分配[10, 71]发挥相同作用，但无需感知输入的二维结构——*i.e.*，YOLOS无需将ViT的输出序列重新解释为二维特征图以进行标签分配。理论上，只要每次处理时始终以相同方式将输入展平为序列，YOLOS能够在不知晓具体空间结构和几何形态的情况下执行任意维度的物体检测。

在更高分辨率下进行微调。在COCO上进行微调时，所有参数均从ImageNet-1k预训练权重初始化，但用于分类和边界框回归的MLP头部以及一百个[DET]标记除外，这些标记是随机初始化的。在微调过程中，图像的分辨率远高于预训练阶段。我们保持补丁大小 P 不变，*i.e.*、 $P \times P = 16 \times 16$ ，这导致了更长的有效序列长度。虽然ViT能够处理任意输入序列长度，但位置嵌入需要适应不同长度的更长输入序列。我们实时²对预训练的位置嵌入进行二维插值。

归纳偏置。我们精心设计了YOLOS架构，以最小化额外归纳偏置的引入。ViT固有的归纳偏置源于网络起始部分的图像块提取以及位置嵌入的分辨率调整[21]。除此之外，YOLOS在ViT³基础上未添加任何非退化（*e.g.*、 3×3 或其他非 1×1 ）卷积操作。从表征学习的角度，我们选择使用[DET]标记来绑定对象以进行最终预测，从而避免引入额外的二维归纳偏置及任务特定启发式方法。受现代CNN架构启发的性能导向设计——如金字塔特征层级、二维局部空间注意力以及区域池化操作——均未采用。所有这些努力旨在以纯粹的序列到序列方式，在最小化依赖输入空间结构与几何先验知识的前提下，精确揭示预训练Transformer从图像识别到目标检测的通用性与可迁移性。

与DETR的对比。YOLOS的设计深受DETR [10] 的启发：YOLOS遵循DETR采用 $\{v^*\}$ 标记作为物体表征的代理，以避免在标签分配过程中引入关于二维结构的归纳偏置和任务先验知识，且YOLOS的优化方式与DETR类似。

与此同时，两种模型也存在一些关键差异：(1) DETR采用Transformer编码器-解码器架构，而YOLOS选择了仅编码器的Transformer架构。(2) DETR仅对其CNN骨干网络进行预训练，而Transformer编码器和解码器则从随机初始化开始训练；YOLOS则天然继承了任何预训练标准ViT的表征能力。(3) DETR在编码后的图像特征与目标查询之间应用交叉注意力，并在每个解码层通过辅助解码损失进行深度监督；而YOLOS在每个编码层始终只关注单一序列，在操作层面不区分 $\{v^*\}$ 令牌和 $\{v^*\}$ 令牌。两者的量化比较详见第3.4节。

3 实验

3.1 设置

预训练。我们使用Touvron等人[57]提出的数据高效训练策略，在ImageNet-1k[51]数据集上对所有YOLOS/ViT模型进行预训练。参数采用截断正态分布初始化，并使用AdamW[40]进行优化。学习率和批量大小分别为 1×10^{-3} 和1024。学习率衰减采用余弦退火策略，权重衰减为0.05。数据增强采用timm库[64]实现的Rand-Augment[14]和随机擦除[69]。正则化方法采用随机深度[32]、Mixup[68]和Cutmix[66]。

²位置嵌入的配置细节详见附录。³我们认为，说Transformer没有卷积是不准确的。Transformer中的所有线性投影层都等价于点积或 1×1 卷积，具有稀疏连接、参数共享和等价表示特性，相比全连接设计中“全对全”交互（其归纳偏置更弱）[5, 24]，这能大幅提升计算效率。

Fine-tuning. We fine-tune all YOLOS models on COCO object detection benchmark [36] in a similar way as Carion et al. [10]. All the parameters are initialized from ImageNet-1k pre-trained weights except for the MLP heads for classification & bounding box regression as well as one hundred [DET] tokens, which are randomly initialized. We train YOLOS on a single node with $8 \times 12\text{G}$ GPUs. The learning rate and batch sizes are 2.5×10^{-5} and 8 respectively. The learning rate decay is cosine and the weight decay is 1×10^{-4} .

As for data augmentation, we use multi-scale augmentation, resizing the input images such that the shortest side is at least 256 and at most 608 pixels while the longest at most 864 for tiny models. For small and base models, we resize the input images such that the shortest side is at least 480 and at most 800 pixels while the longest at most 1333. We also apply random crop augmentations during training following Carion et al. [10]. The number of [DET] tokens are 100 and we keep the loss function as well as loss weights the same as DETR, while we don’t apply dropout [54] or stochastic depth during fine-tuning since we find these regularization methods hurt performance.

Model Variants. With available computational resources, we study several YOLOS variants. Detailed configurations are summarized in Tab. 1. The input patch size for all models is 16×16 . YOLOS-Ti (Tiny), -S (Small), and -B (Base) directly correspond to DeiT-Ti, -S, and -B [57]. From the model scaling perspective [20, 56, 60], the small and base models of YOLOS / DeiT can be seen as performing width scaling (w) [30, 67] on the corresponding tiny model.

Model	DeiT [57] Model	Layers (Depth)	Embed. Dim. (Width)	Pre-train Resolution	Heads	Params.	FLOPs	$\frac{f(\text{Lin.})}{f(\text{Att.})}$
YOLOS-Ti	DeiT-Ti		192		3	5.7 M	1.2 G	5.9
YOLOS-S	DeiT-S	12	384	224	6	22.1 M	4.5 G	11.8
YOLOS-B	DeiT-B		768		12	86.4 M	17.6 G	23.5
YOLOS-S (<i>dwr</i>)	–	19	240	272	6	13.7 M	4.6 G	5.0
YOLOS-S (<i>dwr</i>)	–	14	330	240	6	19.0 M	4.6 G	8.8

Table 1: Variants of YOLOS. “*dwr*” and “*dwr*” refer to uniform compound model scaling and fast model scaling, respectively. The “*dwr*” and “*dwr*” notations are inspired by Dollár et al. [20]. Note that all the numbers listed are for pre-training, which could change during fine-tuning, *e.g.*, the resolution and FLOPs.

Besides, we investigate two other model scaling strategies which proved to be effective in CNNs. The first one is uniform compound scaling (*dwr*) [20, 56]. In this case, the scaling is uniform w.r.t. FLOPs along all model dimensions (*i.e.*, width (w), depth (d) and resolution (r)). The second one is fast scaling (*dwr*) [20] that encourages primarily scaling model width (w), while scaling depth (d) and resolution (r) to a lesser extent w.r.t. FLOPs. During the ImageNet-1k pre-training phase, we apply *dwr* and *dwr* scaling to DeiT-Ti ($\sim 1.2\text{G}$ FLOPs) and scale the model to $\sim 4.5\text{G}$ FLOPs to align with the computations of DeiT-S. Larger models are left for future work.

For canonical CNN architectures, the model complexity or FLOPs (f) are proportional to dw^2r^2 [20]. Formally, $f(\text{CNN}) \propto dw^2r^2$. Different from CNN, there are two kinds of operations that contribute to the FLOPs of ViT. The first one is the linear projection (Lin.) or point-wise convolution, which fuses the information across different channels point-wisely via learnable parameters. The complexity is $f(\text{Lin.}) \propto dw^2r^2$, which is the same as $f(\text{CNN})$. The second one is the spatial attention (Att.), which aggregates the spatial information depth-wisely via computed attention weights. The complexity is $f(\text{Att.}) \propto dwr^4$, which grows quadratically with the input sequence length or number of pixels.

Note that the available scaling strategies are designed for architectures with complexity $f \propto dw^2r^2$, so theoretically the *dwr* as well as *dwr* model scaling are not directly applicable to ViT. However, during pre-training phase the resolution is relatively low, therefore $f(\text{Lin.})$ dominates the FLOPs ($\frac{f(\text{Lin.})}{f(\text{Att.})} > 5$). Our experiments indicate that some model scaling properties of ViT are consistent with CNNs when $\frac{f(\text{Lin.})}{f(\text{Att.})}$ is large.

3.2 The Effects of Pre-training

We study the effects of different pre-training strategies (both label-supervised and self-supervised) when transferring ViT (DeiT-Ti and DeiT-S) from ImageNet-1k to the COCO object detection benchmark via YOLOS. For object detection, the input shorter size is 512 for tiny models and is 800 for small models during inference. The results are shown in Tab. 2 and Tab. 3.

微调。我们以与Carion等人[10]类似的方式，在COCO目标检测基准[36]上对所有YOLOS模型进行微调。除用于分类和边界框回归的MLP头部以及一百个[DET]标记（这些标记是随机初始化的）外，所有参数均从ImageNet-1k预训练权重初始化。我们在单个节点上使用8块 $\times$ 12G GPU训练YOLOS。学习率和批量大小分别为 2.5×10^{-5} 和8。学习率衰减采用余弦方式，权重衰减为 1×10^{-4} 。

在数据增强方面，我们采用多尺度增强方法，对输入图像进行尺寸调整：对于微型模型，最短边至少为256像素、至多为608像素，最长边不超过864像素；对于小型和基础模型，最短边至少为480像素、至多为800像素，最长边不超过1333像素。训练过程中我们还按照Carion等人[10]的方法应用随机裁剪增强。 $\{v^*\}$ 标记的数量为100个，损失函数及损失权重保持与DETR一致，而在微调阶段我们不使用dropout[54]或随机深度正则化，因为我们发现这些正则化方法会损害模型性能。

模型变体。利用可用的计算资源，我们研究了多个YOLOS变体。详细配置总结于表1。所有模型的输入补丁大小均为 16×16 。YOLOS-Ti（微型）、-S（小型）和-B（基础型）直接对应DeiT-Ti、-S和-B[57]。从模型缩放的角度来看[20,56,60]，YOLOS/DeiT的小型 and 基础型模型可视为对相应微型模型进行了宽度缩放 w [30,67]。

Model	DeiT [57] Model	Layers (Depth)	Embed. Dim. (Width)	Pre-train Resolution	Heads	Params.	FLOPs	$\frac{f(\text{Lin.})}{f(\text{Att.})}$
YOLOS-Ti	DeiT-Ti		192		3	5.7 M	1.2 G	5.9
YOLOS-S	DeiT-S	12	384	224	6	22.1 M	4.5 G	11.8
YOLOS-B	DeiT-B		768		12	86.4 M	17.6 G	23.5
YOLOS-S (<i>dwr</i>)	—	19	240	272	6	13.7 M	4.6 G	5.0
YOLOS-S (<i>dwr</i>)	—	14	330	240	6	19.0 M	4.6 G	8.8

表1: YOLOS的变体。“*dwr*”和“v14”分别指统一的复合模型缩放和快速模型缩放。“*dwr*”和“v16”的命名灵感来自Dollár等人[20]。请注意，所列数字均为预训练阶段的数据，在微调过程中可能发生变化，e.g., 包括分辨率和浮点运算次数。

此外，我们研究了另外两种在CNN中被证明有效的模型缩放策略。第一种是均匀复合缩放（*dwr*）[20, 56]。在这种情况下，缩放在所有模型维度（i.e.、宽度（ w ）、深度（ d ）和分辨率（ r ））上相对于FLOPs是均匀的。第二种是快速缩放（*dwr*）[20]，它主要鼓励缩放模型宽度（ w ），同时相对FLOPs较小程度地缩放深度（ d ）和分辨率（ r ）。在ImageNet-1k预训练阶段，我们对DeiT-Ti（ ~ 1.2 G FLOPs）应用*dwr*和*dwr*缩放，并将模型缩放到 ~ 4.5 G FLOPs，以与DeiT-S的计算量对齐。更大的模型留待未来工作。

对于典型的CNN架构，模型复杂度或FLOPs（ f ）与 dw^2r^2 成正比[20]。形式化表示为 $f(\text{CNN}) \propto dw^2r^2$ 。与CNN不同，ViT的FLOPs由两类运算贡献。第一类是线性投影（Lin.）或逐点卷积，它通过可学习参数逐点融合不同通道的信息，其复杂度为 $f(\text{Lin.}) \propto dw^2r^2$ ，与 $f(\text{CNN})$ 相同。第二类是空间注意力（Att.），它通过计算得到的注意力权重沿深度方向聚合空间信息，其复杂度为 $f(\text{Att.}) \propto dwr^4$ ，随输入序列长度或像素数量呈二次方增长。

需要注意的是，可用的缩放策略是为复杂度为 $f \propto dw^2r^2$ 的架构设计的，因此理论上*dwr*以及*dwr*的模型缩放并不直接适用于ViT。然而，在预训练阶段分辨率相对较低，因此 $f(\text{Lin.})$ 主导了FLOPs（ $\frac{f(\text{Lin.})}{f(\text{Att.})} > 5$ ）。我们的实验表明，当 $\frac{f(\text{Lin.})}{f(\text{Att.})}$ 较大时，ViT的某些模型缩放特性与CNN保持一致。

3.2 预训练的效果

我们研究了不同预训练策略（包括标签监督和自监督）在将ViT（DeiT-Ti和DeiT-S）通过YOLOS从ImageNet-1k迁移到COCO目标检测基准时的效果。对于目标检测任务，推理时tiny模型的输入短边尺寸为512，small模型为800。结果展示在表2和表3中。

Model	Pre-train Method	Pre-train Epochs	Fine-tune Epochs	Pre-train pFLOPs	Fine-tune pFLOPs	Total pFLOPs	ImNet Top-1	AP
YOLOS-Ti	Rand. Init.	0	600	0	14.2×10^2	14.2×10^2	–	19.7
	Label Sup. [57]	200		3.1×10^2		10.2×10^2	71.2	26.9
	Label Sup. [57]	300	300	4.7×10^2	7.1×10^2	11.8×10^2	72.2	28.7
	Label Sup. (⌘) [57]	300		4.7×10^2		11.8×10^2	74.5	29.7
YOLOS-S	Rand. Init.	0	250	0	5.9×10^3	5.9×10^3	–	20.9
	Label Sup. [57]	100		0.6×10^3		4.1×10^3	74.5	32.0
	Label Sup. [57]	200	150	1.2×10^3	3.5×10^3	4.7×10^3	78.5	36.1
	Label Sup. [57]	300		1.8×10^3		5.3×10^3	79.9	36.1
	Label Sup. (⌘) [57]	300		1.8×10^3		5.3×10^3	81.2	37.2

Table 2: The effects of label-supervised pre-training. “pFLOPs” refers to petaFLOPs ($\times 10^{15}$). “ImNet” refers to ImageNet-1k. “⌘” refers to the distillation method from Touvron et al. [57].

Model	Self Sup. Pre-train Method	Pre-train Epochs	Fine-tune Epochs	Linear Acc.	AP
YOLOS-S	MoCo-v3 [13]	300	150	73.2	33.6
	DINO [11]	800	150	77.0	36.2

Table 3: Study of self-supervised pre-training on YOLOS-S.

Necessity of Pre-training. At least under prevalent transfer learning paradigms [10, 57], the pre-training is necessary in terms of computational efficiency. For both tiny and small models, we find that pre-training on ImageNet-1k saves the total theoretical forward pass computations (total pre-training FLOPs & total fine-tuning FLOPs) compared with training on COCO from random initialization (training from scratch [28]). Models trained from scratch with hundreds of epochs still lag far behind the pre-trained ViT even if given more total FLOPs budgets. This seems quite different from canonical modern CNN-based detectors, which can catch up with pre-trained counterparts quickly [28].

Label-supervised Pre-training. For supervised pre-training with ImageNet-1k ground truth labels, we find that different-sized models prefer different pre-training schedules: 200 epochs pre-training for YOLOS-Ti still cannot catch up with 300 epochs pre-training even with a 300 epochs fine-tuning schedule, while for the small model 200 epochs pre-training provides feature representations as good as 300 epochs pre-training for transferring to the COCO object detection benchmark.

With additional transformer-specific distillation (“⌘”) introduced by Touvron et al. [57], the detection performance is further improved by ~ 1 AP for both tiny and small models, in part because exploiting a CNN teacher [47] during pre-training helps ViT adapt to COCO better. It is also promising to directly leverage [DET] tokens to help smaller YOLOS learn from larger YOLOS on COCO during fine-tuning in a similar way as Touvron et al. [57], we leave it for future work.

Self-supervised Pre-training. The success of Transformer in NLP greatly benefits from large-scale self-supervised pre-training [18, 44, 45]. In vision, pioneering works [12, 21] train self-supervised Transformers following the masked auto-encoding paradigm in NLP. Recent works [11, 13] based on siamese networks show intriguing properties as well as excellent transferability to downstream tasks. Here we perform a preliminary transfer learning experiment on YOLOS-S using MoCo-v3 [13] and DINO [11] self-supervised pre-trained ViT weights in Tab. 3.

The transfer learning performance of 800 epochs DINO self-supervised model on COCO object detection is on a par with 300 epochs DeiT label-supervised pre-training, suggesting great potentials of self-supervised pre-training for ViT on challenging object-level recognition tasks. Meanwhile, the transfer learning performance of MoCo-v3 is less satisfactory, in part for the MoCo-v3 weight is heavily under pre-trained. Note that the pre-training epochs of MoCo-v3 are the same as DeiT (300 epochs), which means that there is still a gap between the current state-of-the-art self-supervised pre-training approach and the prevalent label-supervised pre-training approach for YOLOS.

YOLOS as a Transfer Learning Benchmark for ViT. From the above analysis, we conclude that the ImageNet-1k pre-training results cannot precisely reflect the transfer learning performance on COCO object detection. Compared with widely used image recognition transfer learning benchmarks such as CIFAR-10/100 [34], Oxford-IIIT Pets [43] and Oxford Flowers-102 [42], the performance of

Model	Pre-train Method	Pre-train Epochs	Fine-tune Epochs	Pre-train pFLOPs	Fine-tune pFLOPs	Total pFLOPs	ImNet Top-1	AP
YOLOS-Ti	Rand. Init.	0	600	0	14.2×10^2	14.2×10^2	–	19.7
	Label Sup. [57]	200		3.1×10^2		10.2×10^2	71.2	26.9
	Label Sup. [57]	300	300	4.7×10^2	7.1×10^2	11.8×10^2	72.2	28.7
	Label Sup. (🐼) [57]	300		4.7×10^2		11.8×10^2	74.5	29.7
YOLOS-S	Rand. Init.	0	250	0	5.9×10^3	5.9×10^3	–	20.9
	Label Sup. [57]	100		0.6×10^3		4.1×10^3	74.5	32.0
	Label Sup. [57]	200	150	1.2×10^3	3.5×10^3	4.7×10^3	78.5	36.1
	Label Sup. [57]	300		1.8×10^3		5.3×10^3	79.9	36.1
	Label Sup. (🐼) [57]	300		1.8×10^3		5.3×10^3	81.2	37.2

表2: 标签监督预训练的效果。“pFLOPs”指petaFLOPs ($\times 10^{15}$)。“ImNet”指ImageNet-1k。“🐼”指Touvron等人[57]的蒸馏方法。

Model	Self Sup. Pre-train Method	Pre-train Epochs	Fine-tune Epochs	Linear Acc.	AP
YOLOS-S	MoCo-v3 [13]	300	150	73.2	33.6
	DINO [11]	800	150	77.0	36.2

表3: YOLOS-S上的自监督预训练研究。

预训练的必要性。至少在主流的迁移学习范式[10, 57]下, 从计算效率的角度来看, 预训练是必要的。对于微型和小型模型, 我们发现相较于从随机初始化(从头训练[28])在COCO上进行训练, 在ImageNet-1k上进行预训练能够节省总的理论前向传播计算量(总预训练FLOPs和总微调FLOPs)。即使给予更多的总FLOPs预算, 经过数百个周期从头训练的模型仍然远远落后于预训练的ViT。这与经典的现代基于CNN的检测器似乎有很大不同, 后者能够迅速赶上预训练的对应模型[28]。

标签监督的预训练。对于使用ImageNet-1k真实标签进行监督预训练的情况, 我们发现不同规模的模型偏好不同的预训练方案: YOLOS-Ti模型即使采用300轮微调计划, 其200轮预训练仍无法追平300轮预训练的效果; 而对于小型模型, 200轮预训练所提供的特征表示在迁移至COCO目标检测基准时, 其质量与300轮预训练相当。

随着Touvron等人[57]引入了额外的Transformer专用蒸馏方法(“🐼”), 检测性能通过 ~ 1 AP得到进一步提升, 无论是微型还是小型模型均受益。这部分归因于在预训练阶段利用CNN教师模型[47]有助于ViT更好地适应COCO数据集。同样有前景的是, 在微调过程中直接利用[DET]标记, 以类似Touvron等人[57]的方式帮助较小规模的YOLOS从较大规模的YOLOS在COCO数据集上学习, 我们将此方向留待未来研究。

自监督预训练。Transformer在NLP领域的成功很大程度上得益于大规模自监督预训练[18, 44, 45]。在视觉领域, 开创性工作[12, 21]遵循NLP中的掩码自编码范式训练自监督Transformer。近期基于孪生网络的研究[11, 13]展现出引人注目的特性以及出色的下游任务迁移能力。我们在表3中使用MoCo-v3[13]和DINO[11]自监督预训练的ViT权重对YOLOS-S进行了初步迁移学习实验。

经过800个epoch训练的DINO自监督模型在COCO目标检测任务上的迁移学习性能, 与经过300个epoch训练的DeiT标签监督预训练模型相当, 这表明自监督预训练在ViT模型应对具有挑战性的物体级识别任务方面具有巨大潜力。与此同时, MoCo-v3的迁移学习表现则不尽如人意, 部分原因在于MoCo-v3的权重严重欠预训练。值得注意的是, MoCo-v3的预训练epoch数与DeiT相同(均为300个epoch), 这意味着当前最先进的自监督预训练方法与主流的标签监督预训练方法在YOLOS上仍存在差距。

YOLOS作为ViT的迁移学习基准。从上述分析中, 我们得出结论: ImageNet-1k的预训练结果无法准确反映在COCO目标检测任务上的迁移学习性能。与广泛使用的图像识别迁移学习基准(如CIFAR-10/100 [34]、Oxford-IIIT Pets [43]和Oxford Flowers-102 [42])相比, 其性能表现

YOLOS on COCO is more sensitive to the pre-train scheme and the performance is far from saturating. Therefore it is reasonable to consider YOLOS as a challenging transfer learning benchmark to evaluate different (label-supervised or self-supervised) pre-training strategies for ViT.

3.3 Pre-training and Transfer Learning Performance of Different Scaled Models

We study the pre-training and the transfer learning performance of different model scaling strategies, *i.e.*, width scaling (w), uniform compound scaling (dwr) and fast scaling (dwr). The models are scaled from $\sim 1.2\text{G}$ to $\sim 4.5\text{G}$ FLOPs regime for pre-training. Detailed model configurations and descriptions are given in Sec. 3.1 and Tab. 1.

We pre-train all the models for 300 epochs on ImageNet-1k with input resolution determined by the corresponding scaling strategies, and then fine-tune these models on COCO for 150 epochs. Few literatures are available for resolution scaling in object detection, where the inputs are usually oblong in shape and the multi-scale augmentation [10, 27] is used as a common practice. Therefore for each model during inference, we select the smallest resolution (*i.e.*, the shorter size) ranging in [480, 800] producing the highest box AP, which is 784 for dwr scaling and 800 for all the others. The results are summarized in Tab. 4.

Scale	Image Classification @ ImageNet-1k				Object Detection @ COCO val			
	FLOPs	$\frac{f(\text{Lin.})}{f(\text{Att.})}$	FPS	Top-1	FLOPs	$\frac{f(\text{Lin.})}{f(\text{Att.})}$	FPS	AP
–	1.2 G	5.9	1315	72.2	81 G	0.28	12.0	29.6
w	4.5 G	11.8	615	79.9	194 G	0.55	5.7	36.1
dwr	4.6 G	5.0	386	80.5	163 G	0.35	4.5	36.2
dwr	4.6 G	8.8	511	80.4	172 G	0.49	5.7	37.6

Table 4: Pre-training and transfer learning performance of different scaled models. FLOPs and FPS data of object detection are measured over the first 100 images of COCO val split during inference following Carion et al. [10]. FPS is measured with batch size 1 on a single 1080Ti GPU.

Pre-training. Both dwr and dwr scaling can improve the accuracy compared with simple w scaling, *i.e.*, the DeiT-S baseline. Other properties of each scaling strategy are also consistent with CNNs [20, 56], *e.g.*, w scaling is the most speed friendly. dwr scaling achieves the strongest accuracy. dwr is nearly as fast as w scaling and is on a par with dwr scaling in accuracy. Perhaps the reason why these CNN model scaling strategies are still applicable to ViT is that during pre-training the linear projection (1×1 convolution) dominates the model computations.

Transfer Learning. The picture changes when transferred to COCO. The input resolution r is much higher so the spatial attention takes over and linear projection part is no longer dominant in terms of FLOPs ($\frac{f(\text{Lin.})}{f(\text{Att.})} \propto \frac{w}{r^2}$). Canonical CNN model scaling recipes do not take spatial attention computations into account. Therefore there is some inconsistency between pre-training and transfer learning performance: Despite being strong on ImageNet-1k, the dwr scaling achieves similar box AP as simple w scaling. Meanwhile, the performance gain from dwr scaling on COCO cannot be clearly explained by the corresponding CNN scaling methodology that does not take $f(\text{Att.}) \propto dwr^4$ into account. The performance inconsistency between pre-training and transfer learning calls for novel model scaling strategies for ViT considering spatial attention complexity.

3.4 Comparisons with CNN-based Object Detectors

In previous sections, we treat YOLOS as a touchstone for the transferability of ViT. In this section, we consider YOLOS as an object detector and we compare YOLOS with some modern CNN detectors.

Comparisons with Tiny-sized CNN Detectors. As shown in Tab. 5, the tiny-sized YOLOS model achieves impressive performance compared with well-established and highly-optimized CNN object detectors. YOLOS-Ti is strong in AP and competitive in FLOPs & FPS even though Transformer is not intentionally designed to optimize these factors. From the model scaling perspective [20, 56, 60], YOLOS-Ti can serve as a promising model scaling start point.

Comparisons with DETR. The relations and differences in model design between YOLOS and DETR are given in Sec. 2.1, here we make quantitative comparisons between the two.

YOLOS在COCO数据集上对预训练方案更为敏感，其性能远未达到饱和。因此，将YOLOS视为一个具有挑战性的迁移学习基准，用于评估ViT的不同（标签监督或自监督）预训练策略是合理的。

3.3 不同规模模型的预训练与迁移学习性能

我们研究了不同模型缩放策略的预训练和迁移学习性能，包括*i.e.*、宽度缩放(w)、均匀复合缩放(dwr)和快速缩放(dwr)。模型在预训练阶段的计算量从~1.2G FLOPs扩展到~4.5G FLOPs范围。详细的模型配置和说明见第3.1节及表1。

我们在ImageNet-1k上以对应缩放策略确定的输入分辨率对所有模型进行了300个周期的预训练，随后在COCO数据集上对这些模型进行了150个周期的微调。目标检测领域关于分辨率缩放的文献较少，该领域输入通常为长条形且普遍采用多尺度增强技术[10, 27]。因此在推理阶段，我们为每个模型在[480, 800]范围内选取能产生最高边界框AP的最小分辨率（*i.e.*，即短边尺寸）： dwr 缩放策略对应784像素，其余所有策略均采用800像素。具体结果汇总于表4。

Scale	Image Classification @ ImageNet-1k				Object Detection @ COCO val			
	FLOPs	$\frac{f(\text{Lin.})}{f(\text{Att.})}$	FPS	Top-1	FLOPs	$\frac{f(\text{Lin.})}{f(\text{Att.})}$	FPS	AP
–	1.2 G	5.9	1315	72.2	81 G	0.28	12.0	29.6
w	4.5 G	11.8	615	79.9	194 G	0.55	5.7	36.1
dwr	4.6 G	5.0	386	80.5	163 G	0.35	4.5	36.2
dwr	4.6 G	8.8	511	80.4	172 G	0.49	5.7	37.6

表4：不同规模模型的预训练和迁移学习性能。目标检测的FLOPs和FPS数据是在推理过程中按照Carion等人[10]的方法，在COCO val分割数据集的前100张图像上测量的。FPS是在单张1080Ti GPU上以批次大小1测量的。

预训练。与简单的 w 缩放相比， dwr 和 dwr 缩放均能提升精度，*i.e.*，即DeiT-S基线。每种缩放策略的其他特性也与CNN一致[20, 56]，*e.g.*， w 缩放对速度最为友好。 dwr 缩放实现了最强的精度。 dwr 的速度几乎与 w 缩放相当，精度则与 dwr 缩放持平。这些CNN模型缩放策略仍适用于ViT的原因或许在于，预训练期间线性投影（ 1×1 卷积）主导了模型计算。

迁移学习。当转移到COCO数据集时，情况发生了变化。输入分辨率 r 更高，因此空间注意力占据主导，线性投影部分在FLOPs（ $\frac{f(\text{Lin.})}{f(\text{Att.})} \propto \frac{w}{r^2}$ ）方面不再占主导地位。经典的CNN模型缩放方法没有考虑空间注意力计算。因此，预训练和迁移学习性能之间存在一些不一致：尽管在ImageNet-1k上表现强劲，但 dwr 缩放实现的边界框AP与简单的 w 缩放相似。同时， dwr 缩放在COCO上的性能提升无法通过不考虑 $f(\text{Att.}) \propto dwr^4$ 的相应CNN缩放方法论来明确解释。预训练和迁移学习之间的性能不一致，要求针对ViT提出考虑空间注意力复杂度的新型模型缩放策略。

3.4 与基于CNN的目标检测器比较

在前面的章节中，我们将YOLOS视为ViT可迁移性的试金石。在本节中，我们将YOLOS视为目标检测器，并将其与一些现代CNN检测器进行比较。

与微型CNN检测器的比较。如表5所示，与成熟且高度优化的CNN目标检测器相比，微型YOLOS模型取得了令人印象深刻的性能。尽管Transformer并非专门为优化这些因素而设计，但YOLOS-Ti在AP指标上表现强劲，在FLOPs和FPS方面也具备竞争力。从模型缩放的角度来看[20, 56, 60]，YOLOS-Ti可以作为一个有前景的模型缩放起点。

与DETR的对比。YOLOS与DETR在模型设计上的关联与差异已在第2.1节中说明，此处我们对二者进行定量比较。

Method	Backbone	Size	AP	Params. (M)	FLOPs (G)	FPS
YOLOv3-Tiny [49]	DarkNet [49]	416×416	16.6	8.9	5.6	330
YOLOv4-Tiny [60]	COSA [60]	416×416	21.7	6.1	7.0	371
YOLOS-Ti	DeiT-Ti (🦋) [57]	$256 \times *$	23.1	6.5	3.4	114
CenterNet [70]	ResNet-18 [26]	512×512	28.1	–	–	129
YOLOv4-Tiny (3l) [60]	COSA [60]	320×320	28.7	–	–	252
Def. DETR [72]	FBNet-V3 [15]	$800 \times *$	27.9	12.2	12.3	35
YOLOS-Ti	DeiT-Ti (🦋) [57]	$432 \times *$	28.6	6.5	11.7	84

Table 5: Comparisons with some tiny-sized modern CNN detectors. All models are trained to be fully converged. “Size” refers to input resolution for inference. FLOPs and FPS data are measured over the first 100 images of COCO val split during inference following Carion et al. [10]. FPS is measured with batch size 1 on a single 1080Ti GPU.

Method	Backbone	Epochs	Size	AP	Params. (M)	FLOPs (G)	FPS
Def. DETR [72]	FBNet-V3 [15]	150	$800 \times *$	27.5	12.2	12.3	35
YOLOS-Ti	DeiT-Ti [57]	300	$512 \times *$	28.7	6.5	18.8	60
YOLOS-Ti	DeiT-Ti (🦋) [57]	300	$432 \times *$	28.6	6.5	11.7	84
YOLOS-Ti	DeiT-Ti (🦋) [57]	300	$528 \times *$	30.0	6.5	20.7	51
DETR [10]	ResNet-18-DC5 [26]	150	$800 \times *$	36.9	29	129	7.4
YOLOS-S	DeiT-S [57]		$800 \times *$	36.1	31	194	5.7
YOLOS-S	DeiT-S (🦋) [57]		$800 \times *$	37.2	31	194	5.7
YOLOS-S (dwr)	DeiT-S [57] (dwr Scale [20])		$704 \times *$	37.2	28	123	7.7
YOLOS-S (dwr)	DeiT-S [57] (dwr Scale [20])		$784 \times *$	37.6	28	172	5.7
DETR [10]	ResNet-101-DC5 [26]	150	$800 \times *$	42.5	60	253	5.3
YOLOS-B	DeiT-B (🦋) [57]			42.0	127	538	2.7

Table 6: Comparisons with different DETR models. Tiny-sized models are trained to be fully converged. “Size” refers to input resolution for inference. FLOPs and FPS data are measured over the first 100 images of COCO val split during inference following Carion et al. [10]. FPS is measured with batch size 1 on a single 1080Ti GPU. The “ResNet-18-DC5” implantation is from timm library [64].

As shown in Tab. 6, YOLOS-Ti still performs better than the DETR counterpart, while larger YOLOS models with width scaling become less competitive: YOLOS-S with more computations is 0.8 AP lower compared with a similar-sized DETR model. Even worse, YOLOS-B cannot beat DETR with over $2\times$ parameters and FLOPs. Even though YOLOS-S with *dwr* scaling is able to perform better than the DETR counterpart, the performance gain cannot be clearly explained as discussed in Sec. 3.3.

Interpreting the Results. Although the performance is seemingly discouraging, the numbers are meaningful, as YOLOS is not purposefully designed for better performance, but designed to precisely reveal the transferability of ViT in object detection. *E.g.*, YOLOS-B is directly adopted from the BERT-Base architecture [18] in NLP. This 12 layers, 768 channels Transformer along with its variants have shown impressive performance on a wide range of NLP tasks. We demonstrate that with minimal modifications, this kind of architecture can also be successfully transferred (*i.e.*, $AP = 42.0$) to the challenging COCO object detection benchmark in computer vision from a pure sequence-to-sequence perspective. The minimal modifications from YOLOS exactly reveal the versatility and generality of Transformer.

3.5 Inspecting Detection Tokens

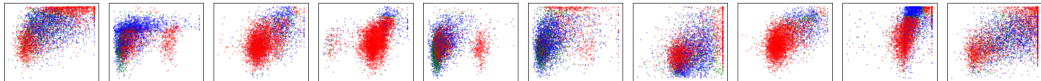


Figure 2: Visualization of all box predictions on all images from COCO val split for the first ten [DET] tokens. Each box prediction is represented as a point with the coordinates of its center normalized by each thumbnail image size. The points are color-coded so that blue points corresponds to small objects, green to medium objects and red to large objects. We observe that each [DET] token learns to specialize on certain regions and sizes. The visualization style is inspired by Carion et al. [10].

Method	Backbone	Size	AP	Params. (M)	FLOPs (G)	FPS
YOLOv3-Tiny [49]	DarkNet [49]	416×416	16.6	8.9	5.6	330
YOLOv4-Tiny [60]	COSA [60]	416×416	21.7	6.1	7.0	371
YOLOS-Ti	DeiT-Ti (🦋) [57]	$256 \times *$	23.1	6.5	3.4	114
CenterNet [70]	ResNet-18 [26]	512×512	28.1	–	–	129
YOLOv4-Tiny (3l) [60]	COSA [60]	320×320	28.7	–	–	252
Def. DETR [72]	FBNet-V3 [15]	$800 \times *$	27.9	12.2	12.3	35
YOLOS-Ti	DeiT-Ti (🦋) [57]	$432 \times *$	28.6	6.5	11.7	84

表5：与一些微型现代CNN检测器的比较。所有模型均训练至完全收敛。“尺寸”指推理时的输入分辨率。FLOPs和FPS数据是在推理过程中按照Carion等人[10]的方法，基于COCO val数据集前100张图像测得的。FPS是在单张1080Ti GPU上以批大小1进行测量的。

Method	Backbone	Epochs	Size	AP	Params. (M)	FLOPs (G)	FPS
Def. DETR [72]	FBNet-V3 [15]	150	$800 \times *$	27.5	12.2	12.3	35
YOLOS-Ti	DeiT-Ti [57]	300	$512 \times *$	28.7	6.5	18.8	60
YOLOS-Ti	DeiT-Ti (🦋) [57]	300	$432 \times *$	28.6	6.5	11.7	84
YOLOS-Ti	DeiT-Ti (🦋) [57]	300	$528 \times *$	30.0	6.5	20.7	51
DETR [10]	ResNet-18-DC5 [26]	150	$800 \times *$	36.9	29	129	7.4
YOLOS-S	DeiT-S [57]		$800 \times *$	36.1	31	194	5.7
YOLOS-S	DeiT-S (🦋) [57]		$800 \times *$	37.2	31	194	5.7
YOLOS-S (dwr)	DeiT-S [57] (dwr Scale [20])		$704 \times *$	37.2	28	123	7.7
YOLOS-S (dwr)	DeiT-S [57] (dwr Scale [20])		$784 \times *$	37.6	28	172	5.7
DETR [10]	ResNet-101-DC5 [26]	150	$800 \times *$	42.5	60	253	5.3
YOLOS-B	DeiT-B (🦋) [57]		$800 \times *$	42.0	127	538	2.7

表6：与不同DETR模型的比较。微型模型均训练至完全收敛。“尺寸”指推理时的输入分辨率。FLOPs与FPS数据遵循Carion等人[10]的方法，在推理阶段使用COCO val数据集的前100张图像进行测量。FPS在单张1080Ti GPU上以批次大小1进行测量。“ResNet-18-DC5”实现基于timm库[64]。

如表6所示，YOLOS-Ti仍然比对应的DETR模型表现更好，而通过宽度缩放得到的更大YOLOS模型则竞争力下降：计算量更大的YOLOS-S相比尺寸相似的DETR模型低了0.8 AP。更糟糕的是，YOLOS-B在参数量和FLOPs超过2×的情况下仍无法超越DETR。尽管采用dwr缩放的YOLOS-S能够比对应的DETR模型表现更好，但如第3.3节所讨论的，这种性能提升无法得到明确解释。

结果解读。尽管性能看似令人沮丧，但这些数字具有意义，因为YOLOS并非为追求更优性能而设计，而是旨在精确揭示ViT在目标检测中的可迁移性。E.g., YOLOS-B直接采用了自然语言处理中BERT-Base架构[18]的设计。这种12层、768通道的Transformer及其变体已在众多自然语言处理任务中展现出卓越性能。我们证明，通过极简修改（i.e., AP = 42.0），此类架构也能成功从纯序列到序列的视角迁移至计算机视觉领域极具挑战性的COCO目标检测基准。YOLOS所采用的最小化修改恰恰揭示了Transformer的多功能性与普适性。

3.5 检测令牌检查

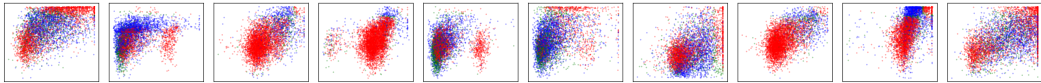


图2：在COCO val分割数据集的所有图像上，对前十个[DET] token的所有边界框预测的可视化。每个边界框预测表示为一个点，其中心坐标已按每个缩略图尺寸归一化。点按颜色编码：蓝色点对应小物体，绿色点对应中等物体，红色点对应大物体。我们观察到每个[DET] token学会专注于特定区域和尺寸。该可视化风格受Carion等人[10]的启发。

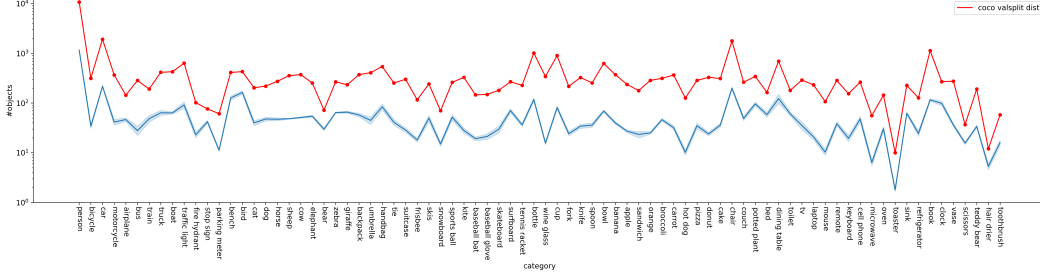


Figure 3: The statistics of all ground truth object categories (the **red** curve) and the statistics of all object category predictions from all [DET] tokens (the **blue** curve) on all images from COCO val split. The error bar of the **blue** curve represents the variability of the preference of different tokens for a given category, which is small. This suggests that different [DET] tokens are category insensitive.

Qualitative Analysis on Detection Tokens. As an object detector, YOLOS uses [DET] tokens to represent detected objects. In general, we find that [DET] tokens are sensitive to object locations and sizes, while insensitive to object categories, as shown in Fig. 2 and Fig. 3.

Quantitative Analysis on Detection Tokens. We give a quantitative analysis on the relation between X = the cosine similarity of [DET] token pairs, and Y = the corresponding predicted bounding box centers ℓ_2 distances. We use the Pearson correlation coefficient $\rho_{X,Y} = \frac{\mathbb{E}[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}$ as a measure of linear correlation between variable X and Y , and we conduct this study on all predicted object pairs within each image in COCO val set averaged by all 5000 images. The result is $\rho_{X,Y} = -0.80$. This means that [DET] tokens that are close to each other (*i.e.*, with high cosine similarity) also lead to mostly nearby predictions (*i.e.*, with short ℓ_2 distances, given $\rho_{X,Y} < 0$).

We also conduct a quantitative study on the relation between X = the cosine similarity of [DET] token pairs, and Y = the corresponding cosine similarity of the output features of the classifier. The result is $\rho_{X,Y} = -0.07$, which is very close to 0. This means that there is no strong linear correlation between these two variables.

Detaching Detection Tokens. To further understand the role [DET] tokens plays, we study impacts caused by detaching the [DET] tokens of YOLOS during training, *i.e.*, we don’t optimize the parameters of the one hundred randomly initialized [DET] tokens. As shown in Tab. 7, detaching the [DET] tokens has a minor impact to AP. These results imply that [DET] tokens mainly serve as the information carrier for the [PATCH] tokens. Similar phenomena are also observed in Fang et al. [22].

Model	[DET] Tokens Config	AP
YOLOS-Ti	Rand. Init. & Learnable	28.7
	Rand. Init. & Detached	28.3
YOLOS-S	Rand. Init. & Learnable	36.1
	Rand. Init. & Detached	36.4

Table 7: Impacts of detaching the [DET] tokens of YOLOS during training.

4 Related Work

Vision Transformer for Object Detection. There has been a lot of interest in combining CNNs with forms of self-attention mechanisms [4] to improve object detection performance [9, 31, 63], while recent works trend towards augmenting Transformer with CNNs (or CNN design). Beal et al. [6] propose to use a pre-trained ViT as the feature extractor for a Faster R-CNN [50] object detector. Despite being effective, they fail to ablate the CNN architectures, region-wise pooling operations [23, 25, 27] as well as hand-crafted components such as dense anchors [50] and NMS. Inspired by modern CNN architecture, some works [39, 59, 62, 65] introduce the pyramidal feature hierarchy and locality to Vision Transformer design, which largely boost the performance in dense prediction tasks including object detection. However, these architectures are performance-oriented and cannot reflect the properties of the canonical or vanilla Vision Transformer [21] that directly inherited from Vaswani et al. [58]. Another series of work, the DETection TRansformer (DETR) families [10, 72], use a random initialized Transformer to encode & decode CNN features for object detection, which does not reveal the transferability of a pre-trained Transformer.

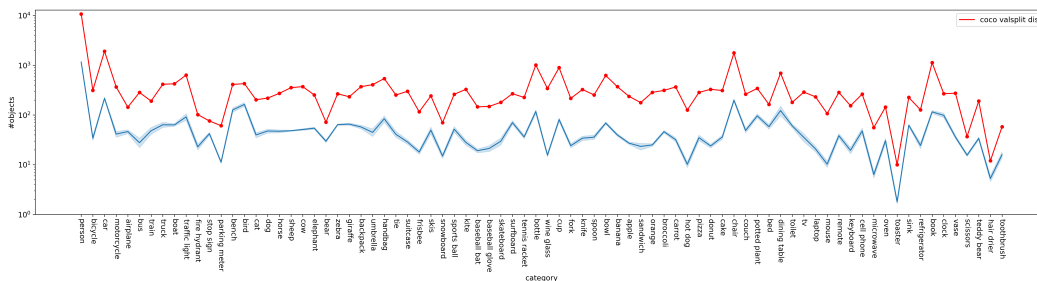


图3: COCO val 分割数据集中所有图像上真实物体类别的统计（红色曲线）与所有 [DET] 标记预测的物体类别统计（蓝色曲线）。蓝色曲线的误差棒表示不同标记对特定类别的偏好变异性，该值较小。这表明不同的 [DET] 标记对类别不敏感。

检测令牌的定性分析。作为目标检测器，YOLOS使用[DET]令牌来表示检测到的目标。总体而言，我们发现[DET]令牌对目标位置和尺寸敏感，而对目标类别不敏感，如图2和图3所示。

检测令牌的定量分析。我们对 X = 余弦相似度与[DET]令牌对之间的关系，以及 Y = 对应预测边界框中心 ℓ_2 的距离进行了定量分析。我们使用皮尔逊相关系数 $\rho_{X,Y} = \frac{\mathbb{E}[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$ 作为变量 X 和 Y 之间线性相关性的度量，并在COCO验证集的每张图像中所有预测对象对上开展研究，结果取5000张图像的平均值。结果为 $\rho_{X,Y} = -0.80$ 。这表明[DET]令牌彼此接近时（*i.e.*即具有高余弦相似度），通常也会产生位置邻近的预测（*i.e.*即具有较短的 ℓ_2 距离，给定 $\rho_{X,Y} < 0$ ）。

我们还对 X = [DET]标记对之间的余弦相似度与 Y = 分类器输出特征的相应余弦相似度之间的关系进行了定量研究。结果为 $\rho_{X,Y} = -0.07$ ，非常接近0。这意味着这两个变量之间不存在强线性相关性。

分离检测令牌。为了进一步理解[DET]令牌所扮演的角色，我们研究了在训练过程中分离YOLOS的[DET]令牌所产生的影响，*i.e.*，即我们不优化一百个随机初始化的[DET]令牌的参数。如表7所示，分离[DET]令牌对AP的影响较小。这些结果表明，[DET]令牌主要充当[PATCH]令牌的信息载体。类似的现象在Fang等人[22]的研究中也被观察到。

Model	[DET] Tokens Config	AP
YOLOS-Ti	Rand. Init. & Learnable	28.7
	Rand. Init. & Detached	28.3
YOLOS-S	Rand. Init. & Learnable	36.1
	Rand. Init. & Detached	36.4

表7: 训练期间分离YOLOS的[DET]标记的影响。

4 相关工作

用于目标检测的视觉Transformer。将CNN与自注意力机制相结合以提升目标检测性能已引起广泛关注[4, 9, 31, 63]，而近期研究趋势转向用CNN（或CNN设计）增强Transformer。Beal等人[6]提出使用预训练ViT作为Faster R-CNN[50]目标检测器的特征提取器。尽管有效，但该方法未对CNN架构、区域池化操作[23, 25, 27]以及人工设计组件（如密集锚框[50]和NMS）进行消融研究。受现代CNN架构启发，部分研究[39, 59, 62, 65]将金字塔特征层次结构与局部性引入视觉Transformer设计，显著提升了包括目标检测在内的密集预测任务性能。然而这些架构以性能为导向，未能体现直接继承自Vaswani等人[58]的经典（原始）视觉Transformer[21]特性。另一系列工作——DEtection TRansformer (DETR) 系列[10, 72]使用随机初始化的Transformer对CNN特征进行编码解码以实现目标检测，但未能揭示预训练Transformer的可迁移性。

UP-DETR [16] is probably the first to study the effects of unsupervised pre-training in the DETR framework, which proposes an “object detection oriented” unsupervised pre-training task tailored for Transformer encoder & decoder in DETR. In this paper, we argue for the characteristics of a pre-trained vanilla ViT in object detection, which is rare in the existing literature.

Pre-training and Fine-tuning of Transformer. The textbook-style usage of Transformer [58] follows a “pre-training & fine-tuning” paradigm. In NLP, Transformer-based models are often pre-trained on large corpora and then fine-tuned for different tasks at hand [18, 44]. In computer vision, Dosovitskiy et al. [21] apply Transformer to image recognition at scale using modern vision transfer learning recipe [33]. They show that a standard Transformer encoder architecture is able to attain excellent results on mid-sized or small image recognition benchmarks (*e.g.*, ImageNet-1k [51], CIFAR-10/100 [34], *etc.*) when pre-trained at sufficient scale (*e.g.*, JFT-300M [55], ImageNet-21k [17]). Touvron et al. [57] achieves competitive Top-1 accuracy by training Transformer on ImageNet-1k only, and is also capable of transferring to smaller datasets [34, 42, 43]. However, existing transfer learning literature of Transformer arrest in image-level recognition and does not touch more complex tasks in vision such as object detection, which is also widely used to benchmark CNNs transferability.

Our work aims to bridge this gap. We study the performance and properties of ViT on the challenging COCO object detection benchmark [36] when pre-trained on the mid-sized ImageNet-1k dataset [51] using different strategies.

5 Discussion

Over recent years, the landscape of computer vision has been drastically transformed by Transformer, especially for recognition tasks [10, 21, 39, 57, 59]. Inspired by modern CNN design, some recent works [39, 59, 62, 65] introduce the pyramidal feature hierarchy as well as locality to vanilla ViT [21], which largely boost the performance in dense recognition tasks including object detection.

We believe there is nothing wrong to make performance-oriented architectural designs for Transformer in vision, as choosing the right inductive biases and priors for target tasks is crucial for model design. However, we are more interested in designing and applying Transformer in vision following the spirit of NLP, *i.e.*, pre-train the *task-agnostic* vanilla Vision Transformer for general visual representation learning first, and then fine-tune or adapt the model on specific target downstream tasks *efficiently*. Current state-of-the-art language models pre-trained on massive amounts of corpora are able to perform few-shot or even zero-shot learning, adapting to new scenarios with few or no labeled data [8, 38, 45, 46]. Meanwhile, prevalent pre-trained computer vision models, including various Vision Transformer variants, still need a lot of supervision to transfer to downstream tasks.

We hope the introduction of Transformer can not only unify NLP and CV in terms of the architecture, but also in terms of the methodology. The proposed YOLOS is able to turn a pre-trained ViT into an object detector with the fewest possible *modifications*, but our ultimate goal is to adapt a pre-trained model to downstream vision tasks with the fewest possible *costs*. YOLOS still needs 150 epochs transfer learning to adapt a pre-trained ViT to perform object detection, and the detection results are far from saturating, indicating the pre-trained representation still has large room for improvement. We encourage the vision community to focus more on the general visual representation learning for the *task-agnostic* vanilla Transformer instead of the *task-oriented* architectural design of ViT. We hope one day, in computer vision, a universal pre-trained visual representation can be easily adapted to various understanding as well as generation tasks with the fewest possible *costs*.

6 Conclusion

In this paper, we have explored the transferability of the vanilla ViT pre-trained on mid-sized ImageNet-1k dataset to the more challenging COCO object detection benchmark. We demonstrate that 2D object detection can be accomplished in a pure sequence-to-sequence manner with minimal additional inductive biases. The performance on COCO is promising, and these preliminary results are meaningful, suggesting the versatility and generality of Transformer to various downstream tasks.

UP-DETR [16] 可能是首个研究无监督预训练在DETR框架中影响的工作，它提出了一种针对DETR中Transformer编码器和解码器定制的“面向目标检测”的无监督预训练任务。在本文中，我们探讨了预训练原始ViT在目标检测中的特性，这在现有文献中较为罕见。

Transformer的预训练与微调。教科书式的Transformer使用方式遵循“预训练与微调”范式。在自然语言处理领域，基于Transformer的模型通常在大规模语料库上进行预训练，随后针对不同任务进行微调。在计算机视觉领域，Dosovitskiy等人采用现代视觉迁移学习方案，将Transformer应用于大规模图像识别。他们证明，当在足够规模的数据上进行预训练时，标准的Transformer编码器架构能够在中小型图像识别基准上取得优异结果。Touvron等人仅通过在ImageNet-1k上训练Transformer就获得了有竞争力的Top-1准确率，并且能够迁移到更小的数据集。然而，现有Transformer的迁移学习研究多停留在图像级识别任务，尚未触及视觉领域中更复杂的任务（如目标检测），而这类任务也常被用于评估CNN的迁移能力。

我们的工作旨在弥合这一差距。我们研究了在具有挑战性的COCO目标检测基准[36]上，使用不同策略在中等规模的ImageNet-1k数据集[51]上进行预训练时，ViT的性能与特性。

5 讨论

近年来，Transformer彻底改变了计算机视觉的格局，特别是在识别任务方面[10, 21, 39, 57, 59]。受现代CNN设计的启发，近期一些研究[39, 59, 62, 65]为原始ViT[21]引入了金字塔特征层级结构及局部性设计，这显著提升了包括目标检测在内的密集识别任务性能。

我们认为，为视觉领域的Transformer进行以性能为导向的架构设计并无不妥，因为为目标任务选择合适的归纳偏置和先验知识对模型设计至关重要。然而，我们更倾向于遵循自然语言处理的精神来设计和应用视觉Transformer，*i.e.*，首先预训练*task-agnostic*原始视觉Transformer以进行通用视觉表示学习，然后在特定的下游目标任务上对模型进行微调或适配*efficiently*。当前在大量语料库上预训练的最先进语言模型能够进行少样本甚至零样本学习，仅需少量或无需标注数据即可适应新场景[8, 38, 45, 46]。与此同时，包括各种视觉Transformer变体在内的主流预训练计算机视觉模型，在迁移到下游任务时仍需要大量监督信息。

我们希望Transformer的引入不仅能统一NLP和CV的架构，也能在方法论层面实现统一。提出的YOLOS能够以最少的*modifications*将预训练的ViT转化为目标检测器，但我们的最终目标是以最少的*costs*使预训练模型适应下游视觉任务。YOLOS仍需要150个周期的迁移学习来使预训练ViT适应目标检测，且检测结果远未饱和，这表明预训练表征仍有巨大改进空间。我们鼓励视觉社区更关注针对*task-agnostic*原始Transformer的通用视觉表征学习，而非ViT的*task-oriented*架构设计。我们希望有一天在计算机视觉领域，一个通用的预训练视觉表征能够以最少的*costs*轻松适应各种理解与生成任务。

6 结论

本文探讨了在中等规模的ImageNet-1k数据集上预训练的原始ViT模型向更具挑战性的COCO目标检测基准的可迁移性。我们证明了二维目标检测能以纯序列到序列的方式实现，仅需引入极少的额外归纳偏置。模型在COCO数据集上表现出色，这些初步成果具有重要意义，表明Transformer架构具备应对各类下游任务的通用性与泛化能力。

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Checklist

1. For all authors...
 - (a) Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope? [\[Yes\]](#) See Sec. 3.2, Sec. 3.3 and Sec. 3.4.
 - (b) Did you describe the limitations of your work? [\[Yes\]](#) See Sec. 3.2, Sec. 3.3 and Sec. 3.4.
 - (c) Did you discuss any potential negative societal impacts of your work? [\[Yes\]](#) See Sec. 3.2 and Tab. 2 for the total theoretical computations analysis.
 - (d) Have you read the ethics review guidelines and ensured that your paper conforms to them? [\[Yes\]](#) .
2. If you are including theoretical results...
 - (a) Did you state the full set of assumptions of all theoretical results? [\[N/A\]](#)
 - (b) Did you include complete proofs of all theoretical results? [\[N/A\]](#)
3. If you ran experiments...
 - (a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? [\[Yes\]](#) We include them in the supplemental material.
 - (b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)? [\[Yes\]](#) See Sec. 3.1.
 - (c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? [\[Yes\]](#) See Appendix.
 - (d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? [\[Yes\]](#) See Sec. 3.1 and Sec. 3.2.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets...
 - (a) If your work uses existing assets, did you cite the creators? [\[Yes\]](#)
 - (b) Did you mention the license of the assets? [\[Yes\]](#) In the supplementary material.
 - (c) Did you include any new assets either in the supplemental material or as a URL? [\[Yes\]](#) In the supplementary material.
 - (d) Did you discuss whether and how consent was obtained from people whose data you’re using/curating? [\[N/A\]](#)
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? [\[N/A\]](#)
5. If you used crowdsourcing or conducted research with human subjects...
 - (a) Did you include the full text of instructions given to participants and screenshots, if applicable? [\[N/A\]](#)
 - (b) Did you describe any potential participant risks, with links to Institutional Review Board (IRB) approvals, if applicable? [\[N/A\]](#)
 - (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? [\[N/A\]](#)

检查清单

1. 对于所有作者...

(a) 摘要和引言中提出的主要主张是否准确反映了论文的贡献与范围? [是] 参见第3.2节、第3.3节及第3.4节。 (b) 是否描述了工作的局限性? [是] 参见第3.2节、第3.3节及第3.4节。 (c) 是否讨论了工作可能带来的负面社会影响? [是] 关于整体理论计算分析, 参见第3.2节及表2。 (d) 是否阅读了伦理审查指南并确保论文符合其要求? [是]

2. 若您包含理论结果...

(a) 你是否阐明了所有理论结果的完整假设条件? [不适用] (b) 你是否包含了所有理论结果的完整证明? [不适用]

3. 如果您进行了实验...

(a) 是否包含了重现主要实验结果所需的代码、数据和说明(可在补充材料或通过URL获取)? [是] 我们已将其包含在补充材料中。 (b) 是否详细说明了所有训练细节(例如数据划分、超参数及其选择依据)? [是] 参见第3.1节。 (c) 是否报告了误差范围(例如基于多次实验的随机种子波动)? [是] 参见附录。 (d) 是否说明了计算总量及所用资源类型(例如GPU型号、内部集群或云服务商)? [是] 参见第3.1节和第3.2节。

4. 如果您正在使用现有资产(例如代码、数据、模型)或整理/发布新资产...

(a) 若您的工作使用了现有资源, 是否已注明创作者? [是] (b) 是否提及了资源的许可协议? [是] 在补充材料中。 (c) 是否在补充材料或通过URL提供了任何新资源? [是] 在补充材料中。 (d) 是否讨论了从数据提供者/整理者处获取同意的方式及情况? [不适用] (e) 是否讨论了所用/整理的数据是否包含个人身份信息或冒犯性内容? [不适用]

5. 如果您使用了众包或进行了涉及人类受试者的研究...

(a) 是否包含了提供给参与者的完整说明文本及适用情况下的截图? [不适用] (b) 是否描述了任何潜在的参与者风险, 并附上机构审查委员会(IRB)批准链接(如适用)? [不适用] (c) 是否注明了支付给参与者的估计时薪及参与者补偿总支出? [不适用]